Pr7: Pizza parlor accepting maximum M orders. Orders are served on a first come first served basis. Queues are frequently used in computer programming, and a typical example is the creation of a job queue by an operating system. If the operating system does not use priorities, then the jobs are processed in the order they enter the system. Write a program for simulating job queue. Write functions to add jobs and delete jobs from the queue.

```
#include<iostream>
#include<windows.h>
using namespace std;
const int MAX=5;


class PizzaParlour
{
	int front,rear;
	int orders[MAX];
	public:
		PizzaParlour()
		{
			front=rear=-1;
		}
		bool addOrder(int data);
		void serveOrder();
		void display();
};
bool PizzaParlour::addOrder(int id){
	if(rear==-1)
	{
		front=rear=0;
		orders[rear]=id;
		return true;
	 }
	 else
	 {
		int pos=(rear+1)%MAX;
		if(pos==front)
		{
			cout<<"\nCafe is Full.Please wait.\n";
			return false;
		 }
		 else
		 {
			rear=pos;
```

```
                orders[rear]=id;
                return true;
            }
        }
}

void PizzaParlour::serveOrder()
{
       if(front==-1)
       {
              cout<<"\n No Orders in Cafe.[Cafe is Empty)\n";
              return;
        }
        else
        {
              cout<<"\n Order No. "<<orders[front]<<" is
processed.\n";
              if(front==rear) //only one order
              {
                     front=rear=-1;
              }
              else
              {
                     front=(front+1)%MAX;
              }
        }
}

void PizzaParlour::display()
{
       int i=0;
       if(front==-1)
       {
              cout<<"\nCafe is Empty.No orders.\n";
              return;
        }
        else
        {
              cout<<"Order Id's: \n";
              for(i=front;i!=rear;i=((i+1)%MAX))
              {
                     cout<<orders[i]<<"  ";
              }
```

```
                cout<<orders[rear];
        }
}
void intro()
{       char name[50]={"\n Vaibhav Cafe \n"};
                for(int i=0;name[i]!='\0';i++)
        {
                Sleep(50);
                cout<<name[i];

        }
}
int main()
{
        int ch,id=0;

        PizzaParlour q;


        do
        {
                cout<<"\n------------------";
        intro();
                cout<<"------------------";
                cout<<"\n****Menu*****\n";
                cout<<"1. Accept order\n";
                cout<<"2. Serve order\n";
                cout<<"3. Display orders\n";
                cout<<"4. Exit";


                cout<<"\nChoice: ";
                cin>>ch;


                switch(ch)
                {
                case 1:
                                id++;
                                if(q.addOrder(id))
                                {
                                        cout<<"Thank you for
order.Order id is : "<<id;
```

```
                        }
                        else
                        {
                                id--;
                        }
                        break;


                case 2: q.serveOrder();
                        break;


                case 3: q.display();
                        break;
                }
        }while(ch!=4);
        cout<<"\nThank You.Keep Visiting.";


}
```